

MLFF Architecture Revision 73

SIZE-HALVE2 fixed-eight qualified-only 3/10/30 funnel

mdstats project

2026-08-16

Revision 73

Revision 73 implements SIZE-HALVE2 as a separate pre-migration authority. It freezes exactly eight possible target sizes from 128 through 16,384, admits only independently hard-qualified MV sizes to training, requires at least four qualifiers, and implements the exact $q \rightarrow \min(q, 4) \rightarrow 2 \rightarrow 1$ 3/10/30 evidence transitions.

The authority inherits PERF-P2R exact continuation: epoch-10 evidence authenticates epoch-3 checkpoint/optimizer/RNG state, epoch-30 evidence authenticates epoch-10 state, and all candidates share foundation, evaluation-role, TRAIN2-policy, training-run, and schedule identity, exact normalized schedule progress, and increasing exposure. Largest-boundary tie protection remains limited to early practical-equivalence bands; final selection prefers the smaller equivalent size and reports `nonconverged_at_fixed_ceiling` when the largest qualified boundary remains materially superior.

Campaign preparation persists `size_halve2_plan` after MVQUAL1 and includes it in restart receipts. This gate is intentionally pre-migration: revision-64 TARGET-DATA2C v4, TARGET-DATA2D v2, DATA8 membership, and CuEq policy remain unchanged.

Release: mdstats 0.20.206a0

Dependency-graph schema: 55

Next gate: SIZE-FIDELITY2.